Automation & QA Developer - Assessment Documentation

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GitHub Repository: https://github.com/shamak24/AutomateQATest

1. Overview

This project was completed as part of the Automation & QA Developer assessment. It demonstrates web application quality assurance, API integration, workflow automation using n8n, and automated website uptime monitoring.

2. Task 1 - Web Application QA

Tested the RealWorld Conduit application by exercising its primary user flows and identifying functional and usability issues.

- Documented 10 issues with reproduction steps, expected vs actual behaviour, severity & causes.
- Prepared a structured bug report and root cause analysis of the article deletion confirmation.

3. Task 2 - API Integration Using n8n

Developed a Hacker News automation workflow to retrieve and deliver a daily technology news digest.

Implementation:

- Used the Hacker News public API to retrieve story IDs and story details.
- Processed and ranked stories based on their scores, retaining the top five.
- Enriched the highest-ranked story using the Hacker News user API.
- Implemented conditional branching based on a score threshold of 100.
- Delivered formatted notifications through a Discord webhook with error handling paths.

4. Bonus - Automated Uptime Monitoring

Developed a separate n8n workflow to monitor the tested web application at five-minute intervals.

- Checks HTTP response status and application availability.
- Implements retry logic for failed requests.
- Sends Discord alerts when the application is unavailable.
- Tracks response times and maintains monitoring statistics.
- Generates a daily uptime summary.

5. Technologies Used

n8n, REST APIs, JavaScript, Hacker News API, Discord Webhooks, HTTP Requests, and workflow automation.

6. Deliverables

The public GitHub repository contains the QA report, exported n8n workflow JSON files, workflow screenshots, and project documentation.

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